Structure Transfers, Exponents Do Not:
A Pre-Registered Test of Representational Priors on Natural-Data Grokking

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Abstract

Companion work on algorithmic tasks argued that the grokking delay is causally the time to form task-structured representations, and that injecting structure with a contrastive prior controls whether and when generalization occurs — including a ∼17× flattening of the weight-norm delay-law exponent. We test what survives contact with natural data: MNIST grokking in the Omnigrok regime (MLP, MSE+AdamW, N=1,000, 8× init), with translated views as a label-free prior. **The first result is negative and we lead with it**: in a free-norm race the prior is *slower* on 5/5 seeds (median 1.31×). Trajectory analysis shows why: the auxiliary term freezes the weight norm at ∼91–93 while the baseline's norm decays to 63–68 before it generalizes — the race confounds the prior's structure channel with its known norm side-effect. A pre-registered amendment (predictions, kill criteria, and disclosures committed before any amended run) de-confounds the comparison: **at matched (pinned) norm the prior wins at every norm ≥ 50 on every seed**, with criterion-level ratios 1.44–1.96× at norm 50 and 2.08–2.42× at norm 92, where the baseline fails the pre-registered 0.85 criterion within the 100,000-step budget on 3/4 seeds (budget-attained accuracy 0.836–0.856, still climbing slowly) while the prior passes on 4/4 (0.875–0.887); at a relaxed 0.80 criterion all seeds cross and the prior remains 1.32–1.84× faster. **One pre-registered kill criterion fired and we report it**: the exponent-flattening magnitude does *not* transfer — the fitted slope ratio is 1.17 (a censored lower bound; 1.03 when refit at a relaxed, fully uncensored criterion), versus ∼17× on modular arithmetic; on natural data the prior buys a multiplicative offset, not a slope change. Further pre-registered arms: a cross-example nearest-neighbor prior is null at free norm (2–2 under a pre-stated tally rule) despite 0.81–0.84 label purity; the norm side-effect saturates by λ=0.03, so *no* free-norm dose wins; the supervised prior transfers (2.06× on 5/5 seeds); wrong-structure pairing caps test accuracy at 0.76–0.79 (the whether-dissociation transfers); and plain augmentation of the task loss abolishes the delay outright (17.9× median) — a regularizer effect distinct from representational priming. Practical upshot: free-norm comparisons systematically understate representational priors; evaluate them at matched norm, or control the norm directly and let the prior supply off-optimum robustness.

1 Introduction

Grokking — delayed generalization long after training accuracy saturates [1] — has become a testbed for the question of *what* networks must build internally before they generalize. Companion work [2, 3] argued, on modular arithmetic, that the delay is causally the time to form task-structured representations: a supervised-contrastive prior [4] encoding the task's true equivalence structure controls *whether* generalization occurs against exactly-matched shuffled, wrong-structure, and norm-matched controls; a label-free commutativity prior accelerates more reliably than supervision; the prior's residual failure mode is a weight-norm side-effect [5], and a clamp-value sweep showed structure injection flattens the weight-norm delay-law exponent roughly 17× (plain cross-entropy slows 31× per +10 norm units vs. 1.22× with the prior).

Every one of those results lives on algorithmic tasks where the generalizing circuit's feature family is known [6, 7]. This paper asks the transfer question directly, on the standard natural-data grokking setting — MNIST in the Omnigrok regime [8] — with the natural label-free structure: augmentation views. We report the study in the order it happened, because the order is the finding:

1. **A free-norm race produces a clean negative** (Section 3): the view-positive prior is slower than baseline on 5/5 paired seeds.

2. **The trajectories diagnose a channel confound** (Section 3): the auxiliary term freezes the weight norm at ∼91–93 while the baseline generalizes only after decaying to 63–68 — yet the prior *generalizes at* a norm where the baseline sits at 0.22 test accuracy. The race conflates the structure channel (helpful) with the norm channel (harmful), the exact decomposition of [2].

3. **A pre-registered amendment de-confounds** (Sections 4–5): predictions, kill criteria, and disclosures were committed to the repository *before* any amended run started. At matched (pinned) norm the prior wins on every seed at every norm ≥ 50, as a roughly constant multiplicative offset (uncensored slope ratio 1.03); criterion-level ratios reach 2.08–2.42× at pin 92, where baselines also fail to attain the accuracy bar within budget. One kill criterion fired — the exponent magnitude does not transfer — and we report it as registered.

4. **Content, dose, and probe arms complete the map** (Sections 6–7): a cross-example nearest-neighbor prior is null at free norm; no dose of the view prior wins at free norm because the norm side-effect saturates at tiny λ; class structure forms ∼15–18k steps before behavior moves in the prior arms.

The synthesis (Section 8) is a boundary map. What transfers: the structure channel's causal role, the whether-dissociation of the controls, the norm channel as the dominant clock, and the value of matched-norm evaluation. What does not: the dramatic exponent flattening, and (at free norm) any acceleration at all. What this means in practice: free-norm comparisons — the default way anyone would evaluate an auxiliary loss — can make a genuinely structure-carrying prior look strictly harmful.

2 Setup

Grokking regime. Following Omnigrok [8]: MLP 784−200−200−10 (ReLU), MSE loss on one-hot targets, AdamW (lr $= 10^{-3}$, weight decay 0.01), batch size 200, N=1,000 MNIST training subset, all initial parameters scaled 8×. This yields training accuracy ≥ 0.99 by 600–800 steps and delayed test-set generalization tens of thousands of steps later, as the inflated norm decays. We define t_{fit} (first eval with train accuracy $\geq$ 0.99) and t_{gen} (first eval with test accuracy $\geq$ 0.85), evaluating every 200 steps with a budget of 100,000 steps; runs that never reach criterion are *censored* and conservatively scored at budget wherever a number is needed. Total weight norm $\|W\|$ is the ℓ_2 norm over all model parameters.

Conditions. All auxiliary conditions add $\lambda\,\mathcal{L}_{\text{SupCon}}$ (L_{out} variant, $\tau = 0.1$, linear 64-d projection head, $\lambda = 0.3$ unless stated) on the penultimate representation of two independently translated views (±3px, zero-padded) of each batch image. The task loss is always computed on *clean* inputs

condition	t_{gen} (5 seeds)	vs. baseline (median)	censored	max test acc	$\|W\|$ at t_{gen}
baseline	29,800–39,400	—	0/5	0.87–0.88	63–68
supcon-aug	41,600–46,000	1.31× slower	0/5	0.87–0.89	91–93
supcon-label	15,200–17,400	2.06× faster	0/5	0.92–0.93	98–100
supcon-shufpair	—	blocked	5/5	0.76–0.79	(norm 133–137)
augce	1,800–2,200	17.9× faster	0/5	0.92–0.93	—
norm-pin 23	400–600	delay abolished	0/10	0.91–0.93	23

Table 1: **Free-norm grid (grid4, 35 runs).** The label-free view prior is *slower* on every paired seed; the supervised prior transfers; wrong structure blocks at a 0.76–0.79 accuracy ceiling with norm exploding to 133–137; plain augmentation in the task loss and norm control each abolish the delay. Norm-pin rows pool both pinned arms (baseline and prior), which are indistinguishable at the optimum.

except in the explicit augmentation-reference arm, so auxiliary conditions differ from baseline *only* by the auxiliary term: **baseline**; **augce** (task loss on augmented inputs — the practitioner reference); **supcon-aug** (positives = views of the same image; label-free); **supcon-label** (positives = same class; supervised); **supcon-shufpair** (positives = views of a fixed random partner image; wrong structure, matched machinery); **supcon-nn** (positives = views of the image's pixel-space nearest neighbor under a greedy globally-nearest pairing; label-free *cross-example* structure — couples are constructed exactly as in shufpair, so the two conditions differ only in how couples were chosen; couple label purity is measured and reported, never trained on); and **norm-pin** arms that rescale all model parameters to a fixed $\|W\|$ after every step. Five seeds for the free-norm grid; four seeds ($\{0, 1, 3, 4\}$) for the amendment. All runs are same-seed paired.

3 The free-norm race: a clean negative, and its diagnosis

Table 1 and Figure 1 give the race everyone would run first. The label-free prior loses: t_{gen} 41,600–46,000 vs. baseline 29,800–39,400, slower on 5/5 paired seeds (median 1.31×). If we had stopped here, the conclusion would have been "augmentation-based priors do not transfer."

The trajectories say otherwise. The baseline generalizes only once its norm has decayed to 63–68 — the weight-norm clock of [5, 8] ticking exactly as advertised (at $\|W\| \geq 90$ the baseline's test accuracy is 0.22). The view prior never gets to use that clock: its auxiliary gradient reaches equilibrium with weight decay and pins $\|W\|$ near 91–93 *for the whole run* — the Channel-2 side-effect documented in [2], in its purest form. And yet it generalizes anyway, at a norm where the baseline is at chance-plus. The race grades a structure benefit and a norm handicap as one number, and the handicap is bigger on this task. Three further observations sharpen the picture. The supervised prior *grows* the norm to 98–100 and still wins 2.06× on every seed — when content is right, structure dominates the clock. The wrong-structure control is not slow but *capped*: test accuracy plateaus at 0.76–0.79 while the norm explodes to 133–137 — the whether-dissociation of [2] transfers to natural data. And augce — ordinary data augmentation in the task loss — does not accelerate grokking so much as abolish the phenomenon: t_{gen} 1,800–2,200 (17.9× median), with t_{gen} preceding t_{fit} wherever t_{fit} is reached at all. That is a regularizer reshaping the loss landscape so that high-norm solutions generalize, not a representational prior accelerating a delayed transition; the two must not be conflated when crediting "augmentation."

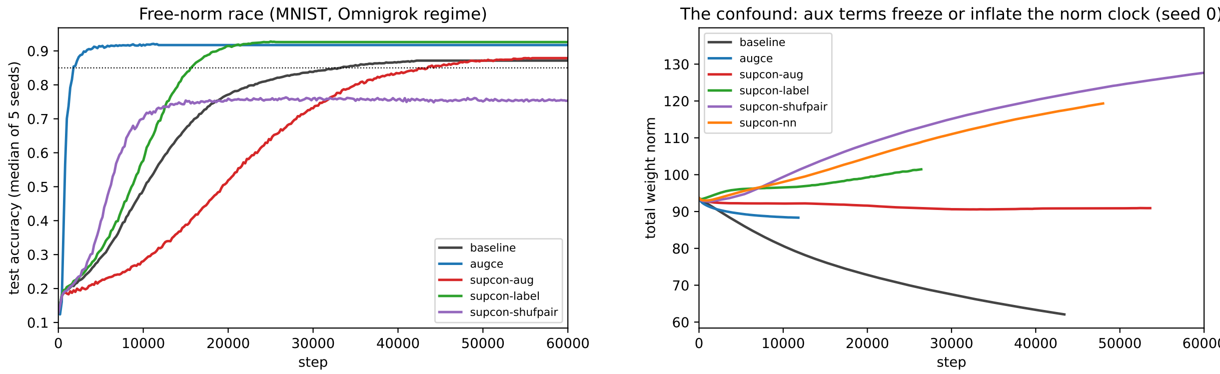

Figure 1: **Left:** free-norm test-accuracy curves (median of 5 seeds). **Right:** the diagnosis — weight-norm trajectories (seed 0). The baseline's norm decays from $\sim$94 and it generalizes only on reaching 63–68. The view prior's auxiliary gradient equilibrates against weight decay and *freezes* the norm near 91–93 for the entire run; the supervised prior grows the norm to 98–100 yet still generalizes 2.06× faster; the wrong-structure control explodes to 133–137 and caps. Norm dynamics and outcomes double-dissociate: content decides whether, norm modulates when.

4 The pre-registered amendment

Diagnosis in hand, the temptation is to re-run comparisons until the prior wins. We instead froze the protocol: predictions, kill criteria, and disclosures were written into the repository and committed (`96f9c0a`) *before any amended run existed*, with verdicts recorded against them at n=2 seeds (`45a1d08`) before the pre-planned extension to n=4, and final verdicts at n=4 (`c9b856e`). Abbreviated (full text in the repository):

- **E2 (matched norm):** pin $\|W\| \in \{35, 50, 65, 80, 92\}$ for baseline and prior ($\|W\|$=23 cells reuse the free grid). *P1*: baseline delay grows steeply with pin, censoring expected by 80–92. *P2* (disclosed as non-blind: the free-norm prior runs sat at $\sim$92 naturally): prior at pin 92 groks near 40k. *P3/K2*: log-delay slope ratio $\geq$ 2, else no meaningful flattening. *K1*: if pinned baseline groks within 1.5× of the prior at 92 on both seeds, there is no matched-norm benefit and the free-norm negative stands in full.

- **E3 (content):** nearest-neighbor couples, purity measured; *P4*: beats same-seed baseline at free norm (conditional on purity $\geq$ 0.8). *K3*: fails on both seeds despite purity $\geq$ 0.8. Extension rule (stated before launching seeds $\{1, 4\}$): verdict = per-seed paired tally over all four seeds, no further seeds afterward.

- **E4 (dose):** $\lambda \in \{0.03, 0.1\}$; *P5*: class-clustering probes rise well before behavior in the prior arms.

- **Disclosures:** the original free-norm negative is reported in full regardless of amendment outcomes (this paper's Section 3); purity is a measured property of MNIST; no amended cell existed at pre-registration time.

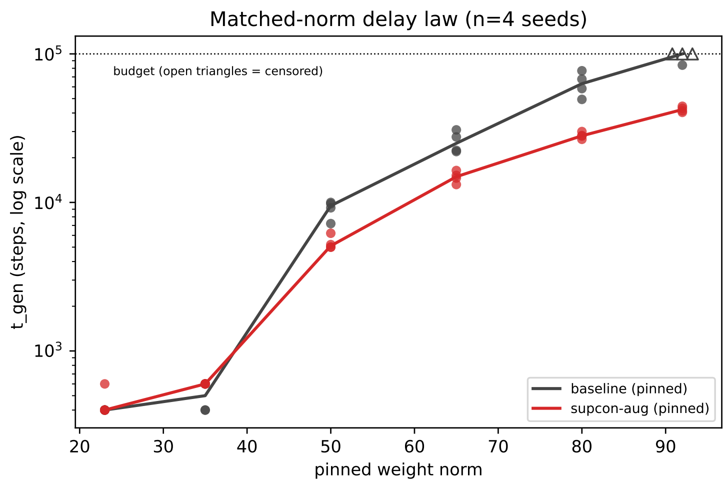

Figure 2: **Matched-norm delay law (n=4 seeds per cell).** t_{gen} vs. pinned $\|W\|$, log scale; open triangles are censored runs at the 100,000-step budget. Below the delay floor (pins 23–35) the arms tie. From pin 50 upward the prior wins on every seed at every norm; criterion-level ratios: 1.44–1.96× at 50, 1.53–2.03× at 65, 1.86–2.57× at 80, 2.08–2.42× at 92 (lower bounds there: 3/4 baseline seeds censor). The two curves are nearly parallel on the log scale — a multiplicative offset, not the algorithmic-task slope collapse.

5 E2: the prior wins wherever there is delay to win; the exponent does not transfer

Figure 2 is the paper's central result. **K1 does not fire**, at any norm, on any seed: pinned at 50 the prior is 1.44–1.96× faster, at 65 1.53–2.03×, at 80 1.86–2.57×, at 92 2.08–2.42× — where the baseline reaches the 0.85 criterion within budget on only one of four seeds (84,000 steps; the others censor at 100,000) while the prior reaches it on all four (40,400–44,400). P1 and P2 are confirmed. The free-norm slowdown of Section 3 is therefore fully attributable to the norm side-effect: remove the handicap by pinning both arms to the same clock, and the structure channel's benefit is present at every operating point where there is delay to win, and largest at the criterion level exactly where the delay law bites hardest (the robustness notes below decompose that growth).

Two honesty notes on the pin-92 cell, both from post-hoc robustness checks (labeled as such; they were not pre-registered). First, censoring at the 0.85 criterion partly reflects budget-attained accuracy, not pure speed: pinned baselines end the budget at 0.836–0.856 test accuracy — still climbing slowly ($\sim$+0.01 per 20k steps at cutoff) — while the prior attains 0.875–0.887. Second, at a relaxed 0.80 criterion every baseline seed does cross (43,000–64,800 vs. the prior's 32,600–36,200), and the prior's advantage is 1.32–1.84× — consistent with the finite-cell ratios at pins 50–80 rather than with the headline $\geq 2.1\times$ criterion-level numbers. The correct summary of pin 92 is therefore: the prior is faster to any fixed accuracy *and* attains visibly higher accuracy within budget; the criterion-level "generalizes vs. does not" framing compounds the two effects.

K2 fires, as pre-registered, and we report it: OLS on $\ln t_{\text{gen}}$ over the ascending branch (pins 50–92, censored cells scored at budget, making the baseline side a lower bound) gives slope 0.0571 per norm unit for baseline (1.77× per +10) vs. 0.0489 for the prior (1.63× per +10): ratio 1.17 (lower bound), far below both the pre-registered bar of 2 and the $\sim$17× observed on modular arithmetic (0.344 vs. 0.020 per unit there). Because the censored cells enter at budget, that fit

seed	baseline t_{gen}	supcon-nn t_{gen}	couple purity	nn final $\|W\|$	outcome
0	33,400	38,000	0.844	119	loss (0.88×)
1	37,600	72,400	0.812	132	loss (0.52×)
3	29,800	28,400	0.830	114	win (1.05×)
4	39,400	38,000	0.822	120	win (1.04×)

Table 2: **E3: cross-example nearest-neighbor prior at free norm.** Tally 2–2 under the pre-stated rule: no evidence of free-norm transfer. Wins are marginal; the seed-1 loss is large (0.52×); purity clears the pre-registered 0.8 bar on every seed.

by itself *fails to confirm* a ratio ≥ 2 rather than excluding one; the relaxed criterion closes the gap. Refit on the same pins and seeds at the 0.80 bar — uncensored in every cell — the slopes are 0.0584 (baseline) vs. 0.0568 (prior): ratio 1.03. The absence of algorithmic-magnitude flattening is therefore an assertion, not a failure to confirm, and no censoring correction to the 0.85 fit could plausibly recover 17×. On natural data the prior shifts the delay-law curve down; it does not change its exponent (the apparent growth of the criterion-level ratios with norm traces to the budget-attained accuracy gap at high pins, per the robustness notes above). The companion's most dramatic quantitative claim is, on this evidence, specific to settings where the generalizing solution is a sparse exact circuit.

6 E3 and E4: content and dose

E3. The nearest-neighbor prior was our best a-priori candidate for a natural-data analogue of the commutativity prior of [3]: label-free *cross-example* structure (couples tie *different* images, mostly within-class at 0.81–0.84 measured purity) rather than same-instance invariance. At free norm it is null: Table 2, tally 2–2 with marginal wins and one large loss. Notably its norm side-effect is the *worst* of any content-bearing prior we test (final $\|W\|$ 114–132, approaching the wrong-structure control's 133–137) — consistent with the confound account: whatever its structure buys at matched norm, at free norm it carries an even heavier clock handicap than the view prior, and breaks even. We did not run its matched-norm sweep and flag it as the natural follow-up; under the confound account its free-norm null is expected rather than exculpatory.

E4. The dose response (Figure 3, left; $\lambda \in \{0, 0.03, 0.1, 0.3\}$): λ=0.03 is break-even with baseline, and every larger dose is monotonically slower. The mechanism is unforgiving: the norm-freeze equilibrium barely moves with λ (final $\|W\| \approx 90$ even at 0.03), so the Channel-2 handicap arrives at full strength before any Channel-1 benefit can pay for it. There is no free lunch at free norm; the confound is not a tuning artifact.

7 P5: structure forms long before behavior

Pre-registered P5 is confirmed in the prior arms with an absolute-threshold operationalization (class cosine gap ≥ 0.09; test accuracy ≥ 0.5): the view prior's probe crosses 17,400–18,200 steps before its behavior (seeds 0 and 3; identically under pin 92), and the nn prior's 8,400–8,800 (Figure 3, right). The baseline is the instructive contrast: its probe crosses *after* its behavior (31,800–35,200 vs. 9,000–9,800) — ungoverned MNIST grokking in this regime does not route through strongly class-clustered penultimate representations; the early structure in the prior arms is prior-made, not

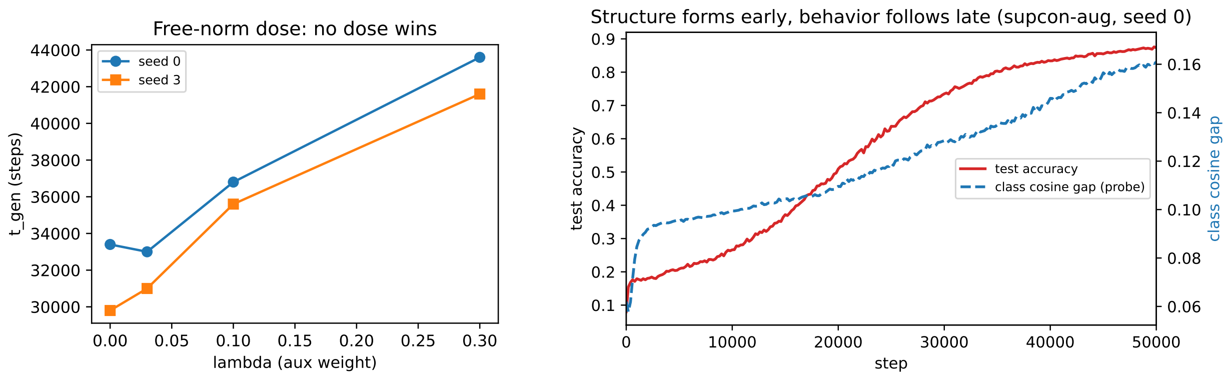

Figure 3: **Left (E4):** free-norm dose response, seeds 0 and 3. λ=0.03 is break-even with baseline and every larger dose is slower; even the tiny dose freezes the norm near 90 (the side-effect saturates before the benefit does), so no dose wins at free norm. **Right (P5):** class cosine gap (probe) vs. test accuracy for the view prior (seed 0): structure is in place by ~1–2k steps; behavior moves at ~16–20k.

a re-description of normal training. (We initially operationalized P5 as 50%-of-max crossing; that degenerates on the baseline's low-amplitude probe curve, and we state the change.)

8 Discussion

What transfers. The causal skeleton of the companion papers survives on natural data: content decides *whether* (supervised prior 2.06× on 5/5; wrong structure capped at 0.76–0.79 with exploding norm), the weight norm is the clock (baseline generalizes precisely as $\|W\|$ decays to 63–68; pinning $\|W\|$ reproduces delay-law growth over more than two orders of magnitude of t_{gen}, from 400–600 steps at pin 23 to beyond the 100,000-step budget at pin 92), the structure channel is real and separable (matched-norm wins on every seed at every pin ≥ 50, growing to 2.08–2.42× where the baseline mostly fails criterion within budget), and structure demonstrably precedes behavior in the prior arms.

What does not. The exponent. On modular arithmetic, structure injection flattened the delay-law slope ~17×; here the slopes are near-parallel (ratio 1.17 censored, 1.03 uncensored). Our working hypothesis, stated as hypothesis: exponent flattening requires the generalizing solution to be a sparse exact circuit whose features the prior can seed directly, as in [6, 7]; MNIST's generalizing solution is a dense statistical classifier, and there a representational head start buys a constant factor, not a different functional dependence on norm. Testing this on tasks that interpolate between the two regimes is the natural next experiment.

Augmentation is two different interventions. The same transformation (translation) enters twice with opposite verdicts: in the task loss (augce) it abolishes the delay (17.9×; generalization precedes memorization), acting as a regularizer that makes high-norm solutions generalize; as contrastive positives it is a representational prior that loses at free norm and wins at matched norm. Papers evaluating "augmentation" in grokking settings should say which intervention they mean; crediting the prior with the regularizer's speed (or damning it with the free-norm race) mistakes the mechanism.

Practical guidance. (1) Free-norm races systematically understate auxiliary representational objectives whose gradients inflate or prop up weight norms — which is most contrastive objectives; evaluate at matched norm, or report norm trajectories alongside speed. (2) On natural data, norm control is first-order (pin 23: 400–600 steps, delay gone) and ordinary augmentation is first-order; a representational prior is the tool for the regime where neither is available at its optimum — its value is off-optimum robustness, the same conclusion the companions reached on algorithmic tasks, now with natural-data evidence. (3) We note candidly that on this small model the auxiliary term costs $\sim$2.5–3$\times$ per step, so the matched-norm step-count advantages of 1.4–2.4$\times$ are at best wall-clock break-even except where the baseline fails criterion entirely; the contribution is mechanistic evidence, not a training recipe.

Relation to prior work. Omnigrok [8] established initialization scale and the norm as the control variables of natural-data grokking; the delay law [5] quantified the norm–delay dependence we lean on throughout; [9, 10] decompose radial vs. tangential learning in ways consistent with our clock/structure split; Grokfast [11] accelerates via gradient filtering and is orthogonal to content injection (the companions found the two do not compose beneficially). Our contribution relative to all of these is the transfer test itself: matched-norm evaluation of a content-bearing prior on natural data, pre-registered, with the negative reported.

9 Limitations and disclosures

One dataset (MNIST), one architecture (a small MLP), one augmentation family (integer translations), n=4–5 seeds per cell: this is a first transfer test, not a survey, and counts are reported instead of p-values wherever cells are small. The pre-registration is repository-native (commit-stamped in our own git history, hashes in Section 4), not third-party; we consider the commits adequate for the claim "predictions preceded results" and note their limits. P2 was disclosed non-blind. K3 as written (both-seed failure at n=2) could not fire on a 1–1 split; the pre-stated four-seed tally rule resolved E3 and we ran no seeds beyond it. The pin-92 robustness checks (budget-attained accuracy; the 0.80 criterion, including the uncensored slope refit) are post-hoc and labeled as such. The 17$\times$ comparator inherits the companion's caveat that its plain-CE slope is itself a censored lower bound. t_{fit} is essentially unaffected by the auxiliary term (600–800 steps across all free-norm content arms), so none of the effects here are memorization-speed artifacts. Greedy nearest-neighbor pairing bounds achievable purity (0.81–0.84); stronger label-free pairings (mutual kNN, learned similarity) and the nn prior's matched-norm sweep are open follow-ups, as is the interpolation hypothesis for the exponent.

10 Reproducibility

This study's internal codename is `paper4` (fourth project phase); it is the third manuscript in the repository, hence the `paper3/` directory — runner and analysis filenames and commit messages use the codename. All numbers in this paper are macros generated from the run artifacts by `paper3/gen_numbers.py` (writing `numbers.tex`) and byte-verified by `paper3/verify_regen.py`. The analysis pipeline is `analysis/analyze_paper4.py`, aggregating `runs/grid4` (35 runs) and `runs/grid4b` (48 runs). Runners are idempotent (`scripts/run_paper4.py` and `scripts/run_paper4b.py`); training code is `src/train_mnist.py`. Pre-registration, two-seed verdicts with the extension rule, and final verdicts are commits `96f9c0a`, `45a1d08`, and `c9b856e`. Hardware: one RTX 3080.

Acknowledgments. This work builds directly on the companion papers' infrastructure and on the weight-norm delay law of Truong et al. [5].

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